

Sigmoid function $\rightarrow$ derivation and prove that

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x) [1 - \sigma(x)]$$

$$\begin{aligned}\sigma'(x) &= \frac{d}{dx} \left(\frac{1}{1+e^{-x}} \right) \\ &= \sigma(x) \left[\frac{1+e^{-x}-1}{1+e^{-x}} \right]\end{aligned}$$

$$\frac{d}{dx} \left(\frac{1}{x} \right) = \frac{d}{dx} (x^{-1})$$

$$-x^{-2} = -\frac{1}{x^2} \quad \left[\text{this is the reciprocal rule of derivatives} \right]$$

now same logic apply here!

$$\frac{d}{dx} \left[\frac{1}{1+e^{-x}} \right] = \frac{d}{dx} \left[(1+e^{-x})^{-1} \right]$$

$$\frac{d}{dx} \left[\frac{1}{1+e^{-x}} \right] = \frac{d}{dx} \left[(1+e^{-x})^{-1} \right]$$

$$\frac{-1}{(1+e^{-x})^2} \frac{d}{dx} (e^{-x}) = \frac{-e^{-x}}{(1+e^{-x})^2} \frac{d}{dx} (-x)$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$

$$\frac{1 \cdot e^{-x}}{(1+e^{-x})(1+e^{-x})} = \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}}$$

$$= \sigma(x) \left[\frac{e^{-x}}{1+e^{-x}} \right]$$

$$= \sigma(x) \left[\frac{1+e^{-x}-1}{1+e^{-x}} \right] = \sigma(x) \left[\frac{1+e^{-x}}{1+e^{-x}} - \frac{1}{1+e^{-x}} \right]$$

$$= \sigma(x) [1 - \sigma(x)]$$

$$\sigma'(x)$$

$$= \cancel{\sigma'(x)} \sigma(x) [1 - \sigma(x)]$$

(Proved)